Spatio-Temporal Kriging of NOAA Daily Maximum Temperature

CM52068: Spatio-Temporal Analytics

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1. Introduction and Data Description

1.1 Data Description

The dataset contains daily weather observations from the US National Oceanic and Atmospheric Administration (NOAA) (Wikle, Zammit-Mangion, and Cressie, 2019). It spans the period 1990 to 1993 and records daily maximum temperatures (Tmax), minimum temperatures (Tmin), dew point temperatures (TDP), and precipitation (Precip) at 137 geographical stations across the United States. Each station is identified by a unique ID and has associated latitude and longitude coordinates. A full description of the variables in the dataset is given in Table 1.

Table 1: Description of variables in the NOAA daily weather dataset.

Variable	Description
`julian`	Julian day number, a continuous day count from a fixed reference date.
`year`	Calendar year of the observation.
`month`	Calendar month of the observation (1–12).
`day`	Day index within the dataset, used as the temporal coordinate t.
`id`	Unique identifier for each weather station.
`z`	Recorded value of the weather variable (units depend on `proc`).
`proc`	The weather variable being recorded (e.g. Tmax, Tmin, TDP, Precip).
`lat`	Latitude of the station in degrees north (°N).
`lon`	Longitude of the station in degrees west (°W), stored as negative values.
`date`	Calendar date of the observation (year-month-day).

1.2 Data Preprocessing

For this analysis, the dataset was filtered to maximum temperature (Tmax) observations during July 1993. The filtered dataset contains 133 unique stations, 31 time steps (days), and 4,122 observations in total. There are zero missing values in the dataset; both explicit `NA` values and potential sentinel values were checked.

Since no missing observations were found after filtering, all $m = 4{,}122$ observations are retained. The dataset is formally defined as:

$$\mathcal{D} = \{(\mathbf{s}_i, t_i, Z_i)\}_{i=1,\ldots,m}, \quad \mathbf{s}_i = (\mathrm{lon}_i, \mathrm{lat}_i), \tag{1}$$

where Z_i is the recorded maximum temperature at spatial location $\mathbf{s}_i$ on day t_i, and m is the total number of known observations.

Station longitudes range from approximately 80.0°W to 100.0°W, and latitudes from approximately 32.1°N to 45.9°N. This covers a broad region of the central and eastern United States, from the southern states (such as Texas and Mississippi) northward to the upper Midwest (such as Wisconsin and Minnesota).

1.3 Visualising the data

Figures 1–3 present three complementary views of the filtered dataset.

Figure 1 shows the distribution of Tmax across all stations and days. The distribution is roughly bimodal, with two peaks centred around 82.6°F–84.4°F and 93.4°F–95.2°F.

Figure 2 displays individual station temperature traces alongside the daily median. The daily mean oscillates around 88.5°F with a total range of approximately 7.9°F (from 84.7°F to 92.6°F). Day-to-day swings average about 1.2°F, with the largest single-day change reaching 3.3°F. This periodic fluctuation reflects weather systems moving through the region. The vertical spread at any given day shows the spatial variability across the station network.

Figure 3 shows the spatial distribution of stations coloured by their mean Tmax. A clear temperature gradient is visible, with the warmest temperatures concentrated in the southwest of the study region and progressively cooler temperatures toward the northeast. Station coverage is reasonably uniform across the region, though some areas are more densely sampled than others.

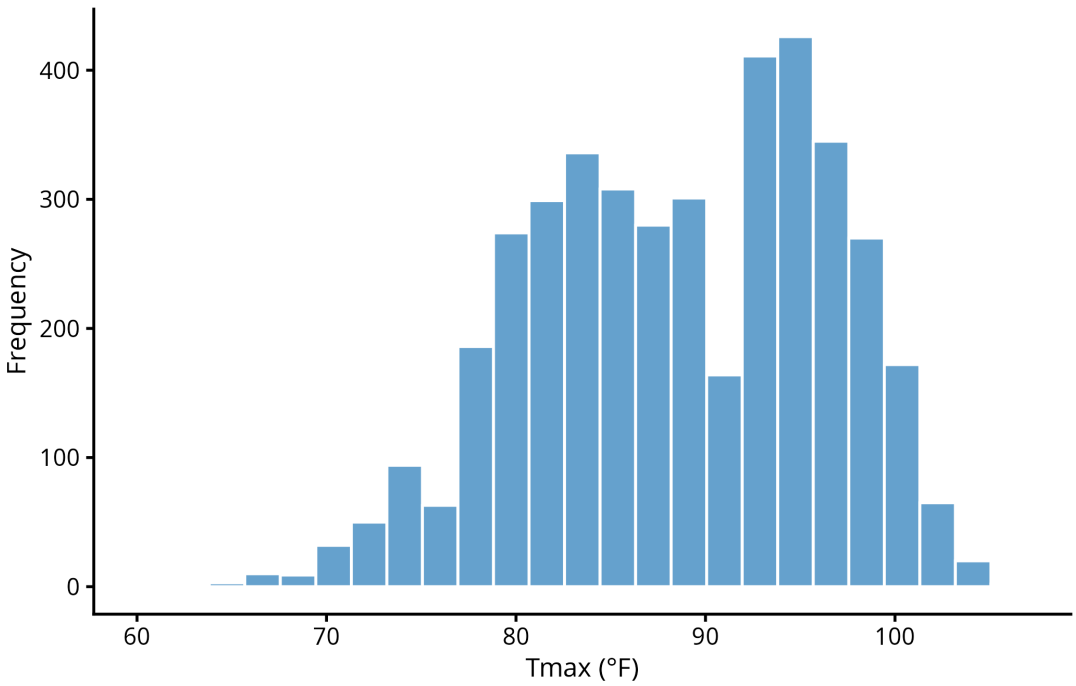

Figure 1: Distribution of daily maximum temperature across all stations in July 1993.

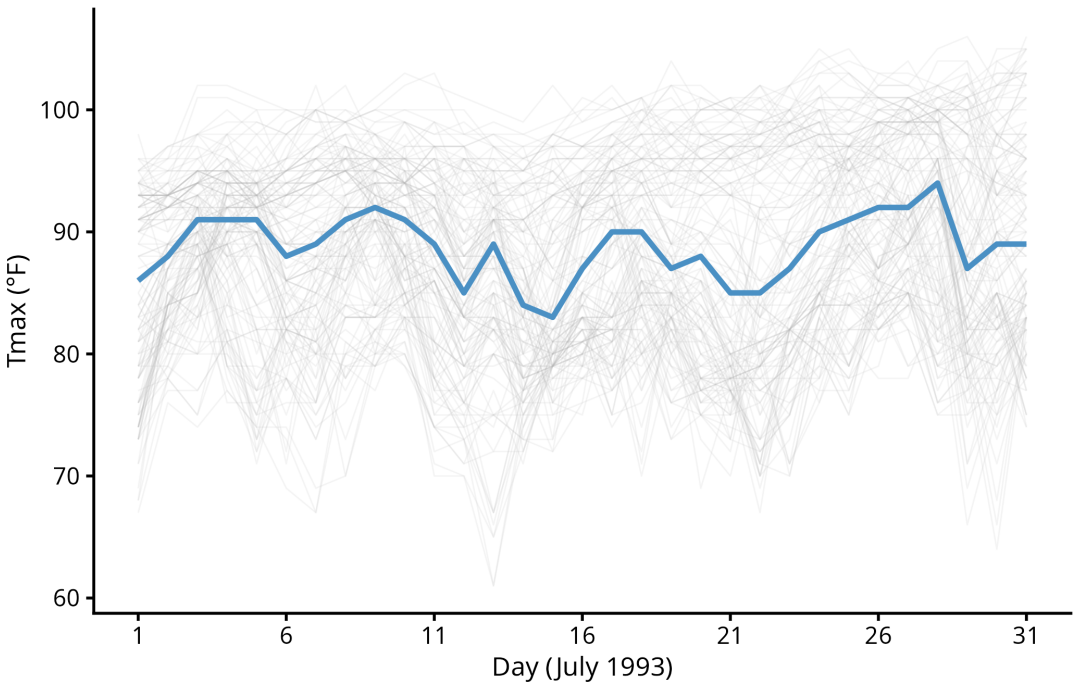

Figure 2: Daily maximum temperature by station (grey) with the median across stations (blue) for July 1993.

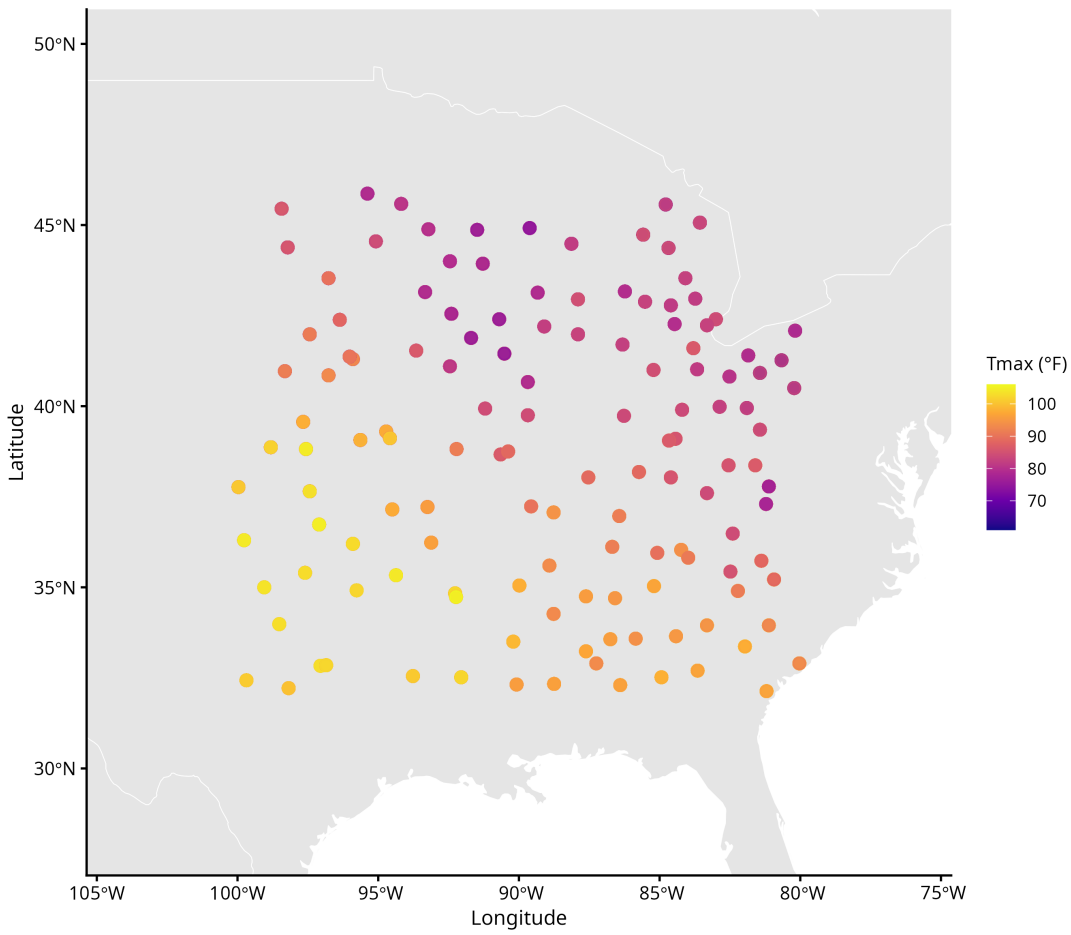

Figure 3: Spatial distribution of maximum temperature observations across station locations in July 1993.

2. Model Descriptions

2.1 Ordinary Least Squares

The OLS model assumes that the maximum temperature Z at a spatial location $\mathbf{s} = (\text{lon}, \text{lat})$ and time t can be decomposed into a deterministic trend and a stochastic noise term:

$$Z(\mathbf{s}, t) = \underbrace{\beta_0 + \beta_1 \,\text{lat} + \beta_2 \,\text{lon} + \beta_3 \, t}_{\text{trend}} + \underbrace{\epsilon(\mathbf{s}, t)}_{\text{noise}}, \tag{2}$$

where β_0 is the intercept, $\beta_1, \beta_2, \beta_3$ are the regression coefficients associated with latitude, longitude, and day respectively, and $\epsilon(\mathbf{s}, t)$ is the residual error. The coefficients are estimated by minimising the sum of squared residuals, yielding the OLS estimates $\hat{\beta}_{\text{OLS}}$.

OLS estimation relies on the assumption that $\epsilon \sim \mathcal{N}(0, \sigma_e^2 \mathbf{I})$, which implies that errors have zero mean, constant variance, and are uncorrelated across observations. In a spatio-temporal setting, the assumption that errors are uncorrelated is particularly difficult to justify. Nearby stations on nearby days tend to share unobserved atmospheric and geographic influences, meaning their residuals are likely to be similar rather than independent. As Wikle, Zammit-Mangion, and Cressie (2019, p.8) note, the standard *iid* error assumption is generally not appropriate when modelling spatio-temporal phenomena, and recognising that errors are statistically dependent in space and time motivates the use of covariance modelling and generalised least squares (GLS). This violation therefore provides the foundation for the kriging approach developed in later sections.

2.2 Covariance Model

The OLS residuals $\mathbf{e} = \mathbf{Z} - \mathbf{X}\hat{\beta}_{\text{OLS}}$ are expected to exhibit spatial and temporal correlation. To capture this dependence structure, we model the covariance between any two observations (s_i, t_i) and (s_j, t_j) using a Gaussian spatio-temporal covariance function (Wikle, Zammit-Mangion, and Cressie, 2019, p.143):

$$C_\theta\big((s_i, t_i), (s_j, t_j)\big) = \sigma^2 \exp\left(-\frac{(\text{lon}_i - \text{lon}_j)^2}{a^2} - \frac{(\text{lat}_i - \text{lat}_j)^2}{b^2} - \frac{(t_i - t_j)^2}{c^2}\right) + \tau^2 \, \mathbf{1}_{\{i=j\}}, \tag{3}$$

where $\theta = (\sigma^2, a, b, c, \tau^2)$ are the model parameters. Each parameter plays a distinct role: σ^2 controls the overall magnitude of the correlated variation; a and b are spatial lengthscale parameters governing how quickly correlation decays in the longitudinal and latitudinal directions respectively; c is the temporal lengthscale controlling the rate of decorrelation over time; and τ^2 is the nugget variance, representing uncorrelated noise such as measurement error or small scale variability that appears only when an observation is compared with itself. Larger values of a, b, or c correspond to stronger dependence over greater distances or longer time spans, while a large τ^2 relative to σ^2 indicates that a substantial proportion of the total variability is due to uncorrelated noise rather than smooth spatial or temporal structure. More precisely, increasing a or b stretches the spatial correlation in the longitudinal or latitudinal direction respectively, so that stations further apart remain more strongly correlated; decreasing them causes correlation to decay more sharply with distance. Similarly, a larger c means observations separated by more days retain stronger temporal correlation, while a small c implies rapid decorrelation over time. Increasing σ^2 raises the overall magnitude of the correlated variation without affecting the rate of decay, and a larger τ^2 relative to σ^2 shifts a greater share of the total variance into uncorrelated noise,

weakening the smooth spatial and temporal structure in the field. The full covariance matrix $\mathbf{C}_\epsilon$ is constructed by evaluating C_θ at every pair of observations in the data. The parameters θ are estimated by maximising the Gaussian log-likelihood (Wikle, Zammit-Mangion, and Cressie, 2019, p.151):

$$\hat{\theta} = \arg\max_{\theta} \left\{ -\frac{1}{2} \left(m \log(2\pi) + \log |\mathbf{C}_\epsilon| + \mathbf{e}^\top \mathbf{C}_\epsilon^{-1} \mathbf{e} \right) \right\}, \tag{4}$$

where $\mathbf{e}$ are the OLS residuals and m is the number of observations. The estimated parameters $\hat{\theta}$ are then used in subsequent sections to construct the covariance matrix for kriging prediction.

2.3 Universal Kriging

With the covariance parameters $\hat{\theta}$ estimated in the previous section, we can now construct the covariance matrix $\mathbf{C}_\epsilon$ and use it to improve both the trend estimation and the predictions at unobserved locations. The first step is to re-estimate the trend coefficients using Generalised Least Squares (GLS). Where OLS assumes $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma_\epsilon^2 \mathbf{I})$, treating all observations as independent, GLS incorporates the estimated covariance structure by assuming $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{C}_\epsilon)$. The GLS estimate of the trend coefficients is given by (Wikle, Zammit-Mangion, and Cressie, 2019, p.142):

$$\hat{\beta}_{\text{GLS}} = (\mathbf{X}^\top \mathbf{C}_\epsilon^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{C}_\epsilon^{-1} \mathbf{Z}. \tag{5}$$

This has the effect of reducing the weight of clusters of correlated observations and giving more influence to spatially or temporally isolated ones, producing more efficient estimates of the trend than OLS. To predict the maximum temperature at an unobserved space-time location $(\mathbf{s}_0, t_0)$, universal kriging combines the GLS trend estimate with a correction based on the residuals at observed locations (Wikle, Zammit-Mangion, and Cressie, 2019, p.142):

$$\hat{Z}(\mathbf{s}_0, t_0) = \mathbf{x}_0^\top \hat{\beta}_{\text{GLS}} + \mathbf{c}_0^\top \mathbf{C}_\epsilon^{-1} (\mathbf{Z} - \mathbf{X}\hat{\beta}_{\text{GLS}}), \tag{6}$$

where $\mathbf{x}_0 = (1, \text{lon}_0, \text{lat}_0, t_0)^\top$ is the vector of covariates at the prediction point, and $\mathbf{c}_0$ is the vector of covariances between the prediction point and all observed locations, computed using the covariance function from Section 2.2 without the nugget term:

$$c_{0,i} = \hat{\sigma}^2 \exp\left(-\frac{(\text{lon}_0 - \text{lon}_i)^2}{\hat{a}^2} - \frac{(\text{lat}_0 - \text{lat}_i)^2}{\hat{b}^2} - \frac{(t_0 - t_i)^2}{\hat{c}^2} \right). \tag{7}$$

The first term of the prediction provides the large-scale trend at the new location, while the second term adjusts this estimate by incorporating information from nearby residuals, weighted by their covariance with the prediction point. If the prediction point is close in space and time to observed stations with large residuals, the prediction is pulled accordingly. If it is far from all observations, $\mathbf{c}_0$ approaches zero and the prediction reverts to the trend alone. Universal kriging also provides a measure of prediction uncertainty. The kriging variance at each prediction point is given by (Wikle, Zammit-Mangion, and Cressie, 2019, p.143):

$$\text{Var}[\hat{Z}(\mathbf{s}_0, t_0) - Z(\mathbf{s}_0, t_0)] = (\hat{\sigma}^2 + \hat{\tau}^2) - \mathbf{c}_0^\top \mathbf{C}_\epsilon^{-1} \mathbf{c}_0 + \mathbf{q}_0^\top (\mathbf{X}^\top \mathbf{C}_\epsilon^{-1} \mathbf{X})^{-1} \mathbf{q}_0, \tag{8}$$

where $\mathbf{q}_0 = \mathbf{x}_0 - \mathbf{X}^\top \mathbf{C}_\epsilon^{-1} \mathbf{c}_0$. The quantity $(\hat{\sigma}^2 + \hat{\tau}^2)$ represents the prior uncertainty in the absence of any data. The term $\mathbf{c}_0^\top \mathbf{C}_\epsilon^{-1} \mathbf{c}_0$ captures the variance reduction gained from nearby observations; the closer and more numerous they are, the larger this quantity becomes and the more it reduces the variance. The remaining term

$\mathbf{q}_0^\top(\mathbf{X}^\top\mathbf{C}_\epsilon^{-1}\mathbf{X})^{-1}\mathbf{q}_0$ accounts for the additional uncertainty arising from having to estimate the trend coefficients rather than knowing them exactly. When covariance parameters are substituted into the kriging equations, the resulting predictor is known as the Empirical Best Linear Unbiased Predictor (Wikle, Zammit-Mangion, and Cressie, 2019, p.151).

3. Evaluation and Comparison

3.1 OLS Results

Table 2 presents the estimated OLS regression coefficients. Latitude ($\hat{\beta}_1 = -1.558$) is by far the strongest predictor of Tmax, with a magnitude considerably larger than those of longitude ($\hat{\beta}_2 = 0.032$) and day ($\hat{\beta}_3 = 0.072$). For every degree further north, Tmax decreases by 1.558°F. The t-value of -83.6 and three-star significance confirm that latitude is the dominant predictor by a large margin, which makes physical sense as latitude governs solar angle and energy input.

Day has a small but highly statistically significant effect ($p < 2 \times 10^{-16}$). Over the full 31 days of July, the cumulative warming amounts to roughly $0.072 \times 30 \approx 2.2$°F, consistent with seasonal heating into midsummer. Longitude has the weakest effect, with a near-zero coefficient that is only marginally significant ($p = 0.012$), suggesting minimal east-west variation in Tmax across the study region.

The model achieves an R^2 of 0.633, meaning the linear trend explains approximately 63% of the total variability in Tmax. The remaining 37%, combined with a residual standard error of 4.698°F, suggests substantial unexplained structure in the residuals. This structure is likely spatially and temporally correlated, which motivated the covariance modelling approach developed in the following sections.

Table 2: Estimated OLS regression coefficients for the Tmax trend model.

Coefficient	**Estimate**	**Std. Error**	***t*-value**	***p*-value**
Intercept ($\hat{\beta}_0$)	150.508	1.334	112.850	$< 2 \times 10^{-16}$
Longitude ($\hat{\beta}_2$)	0.032	0.013	2.518	0.012
Latitude ($\hat{\beta}_1$)	−1.558	0.019	−83.583	$< 2 \times 10^{-16}$
Day ($\hat{\beta}_3$)	0.072	0.008	8.764	$< 2 \times 10^{-16}$
Residual standard error				4.698
R^2				0.633

3.2 Covariance Parameter Estimates

Table 3 reports the estimated covariance parameters $\hat{\theta} = (\hat{\sigma}^2, \hat{a}, \hat{b}, \hat{c}, \hat{\tau}^2)$ obtained by maximising the Gaussian log-likelihood on a random subset of 2000 observations, drawn for computational tractability. Initial parameter values were set to $(16, 5, 3, 1, 4)$ for $\hat{\theta}$ respectively, and were estimated using the L-BFGS-B optimisation method, which supports box constraints to ensure all parameters remain positive throughout the optimisation. The optimiser converged successfully.

The log-likelihood was evaluated efficiently using the Cholesky decomposition of $\mathbf{C}_\epsilon$: the log-determinant was computed as twice the sum of the log-diagonal entries of the Cholesky factor, and the quadratic form $\mathbf{e}^\top\mathbf{C}_\epsilon^{-1}\mathbf{e}$

was evaluated via triangular back-substitution, avoiding explicit matrix inversion. This approach is numerically stable and approximately three times faster per likelihood evaluation than computing the determinant and inverse separately, as confirmed by benchmarking both implementations on the estimation subset.

Table 3: Estimated covariance parameters from maximum likelihood estimation.

Parameter	**Description**	**Estimate**
$\hat{\sigma}^2$	Signal variance	16.362
$\hat{a}$	Longitude lengthscale	5.083
$\hat{b}$	Latitude lengthscale	2.779
$\hat{c}$	Temporal lengthscale	1.140
$\hat{\tau}^2$	Nugget variance	4.035

The signal variance $\hat{\sigma}^2 = 16.362$ and nugget $\hat{\tau}^2 = 4.035$ together make up the total variance of 20.397. Approximately 80% of this is attributable to smooth spatially and temporally correlated variation, with the remaining 20% due to uncorrelated noise such as measurement error or fine-scale variability.

The spatial lengthscales $\hat{a} = 5.083$ (longitude) and $\hat{b} = 2.779$ (latitude) indicate that correlation decays faster in the latitudinal direction than the longitudinal direction. A smaller lengthscale means the covariance drops off more quickly with distance: two stations 3° apart in latitude have a correlation of approximately $\exp(-(3^2/2.779^2)) \approx 0.37$, whereas two stations 3° apart in longitude retain a correlation of approximately $\exp(-(3^2/5.083^2)) \approx 0.70$. The temporal lengthscale $\hat{c} = 1.140$ indicates that correlation between observations drops off within roughly one to two days, meaning that observations more than a few days apart contribute little information to one another.

3.3 Prediction Grid

A 20×20 spatial grid was constructed by taking 20 evenly spaced longitudes in [100 °W, 80°W] and 20 evenly spaced latitudes in [32 °N, 46°N], giving 400 spatial locations. Six days were selected at roughly equal intervals throughout July 1993: days 1, 6, 11, 16, 21, and 26. Combining the spatial grid with these six time steps gives a spatio-temporal prediction grid $\mathcal{G}$ of $400 \times 6 = 2400$ total locations.

3.4 Kriging Predictions and Uncertainty

Like before, the kriging predictions and associated standard errors were computed using the same random subset of 2000 observations used for covariance parameter estimation.

The kriging predictions show a consistent north–south temperature gradient, with the warmest values concentrated in the southern portion of the grid and progressively cooler temperatures toward the north, reflecting the dominant effect of latitude established by the OLS trend.

The uncertainty pattern changes across days as a consequence of the short temporal lengthscale $\hat{c} = 1.140$: observations from days far from the prediction day contribute little information, so the effective local data density varies depending on which day is being predicted. Despite this, there is still some pattern to the data. For example, there is a gradual temperature increase in the southern region where it reaches a maximum at

day 26. The coldest region of the grid starts off as the north-west area, but gradually becomes warmer and the coldest region as time progresses becomes the north-east area.

Uncertainty is lowest (around 2.5°F) in the central and eastern portions of the grid, corresponding to regions with denser station coverage. The highest uncertainty (around 4–4.5°F) appears consistently along the western edge of the grid near 100°W and in the northern corners, where observed stations are sparse and the kriging predictor must rely more heavily on the trend alone.

The standard error maps also appear patchier and less smooth than the prediction maps. This is expected since the predictions are a weighted combination of many observations and the smooth trend surface, which naturally produces a smooth output. The uncertainty, however, depends directly on the distances to the nearest observed stations, which are discretely distributed across the region. Where a station happens to fall close to a grid point, uncertainty drops sharply, creating the patchy appearance. This may explain why the north-east area of the grid has a high error, as it lacks stations in that region due to the border with Canada (see Figure 3).

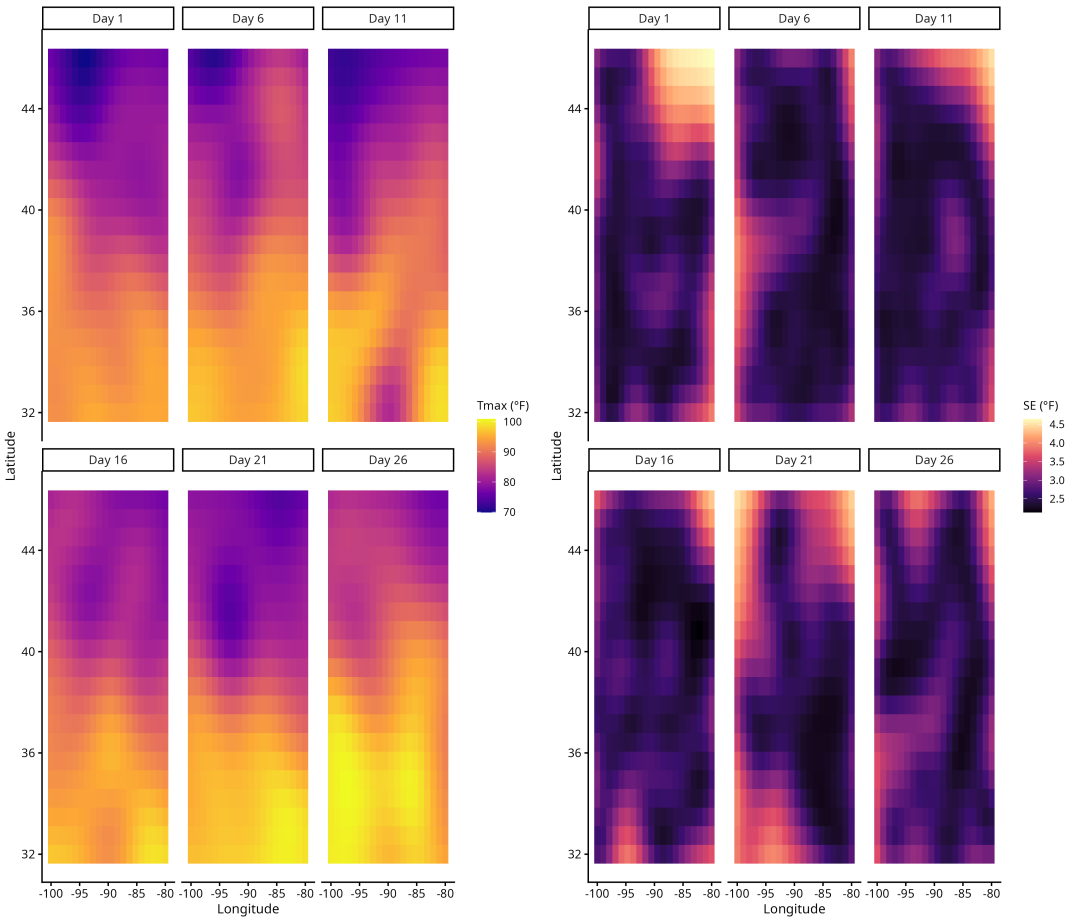

Figure 4: Kriging predictions of Tmax (left) and prediction standard errors (right) over the 20×20 spatial grid for six days in July 1993.

3.5 OLS vs Kriging Comparison

Figure 5 presents the OLS and kriging predictions side by side across the six prediction days.

The OLS predictions produce an almost identical smooth surface across all six days, varying from warm in

the south to cool in the north. There is virtually no east-west variation and almost no change from day to day, reflecting the fact that OLS can only produce predictions from the linear trend in latitude, longitude, and day. The surfaces are essentially indistinguishable across the six days, with only a very subtle uniform shift reflecting the small positive day coefficient.

The kriging predictions tell a very different story. The spatial structure is considerably richer, with localised warm and cool anomalies that shift position and intensity across the month. Features such as the cold spot in the south on Day 11, the cool anomaly in the mid-latitudes on Day 21, and the shift of the cold region from the northwest to the northeast over the course of the month are all captured by kriging but completely absent from OLS. These features arise from the covariance-weighted residual correction term, which pulls predictions toward nearby observed station values. This demonstrates that the covariance model is capturing meaningful residual structure that the linear trend cannot explain.

To quantify this improvement, both models were evaluated on a held-out test set comprising 20% of observations ($n = 824$). OLS achieved a test RMSE of 4.642 °F, while universal kriging achieved 1.565 °F, a reduction of approximately 66.3%. It should be noted that this figure is sensitive to the choice of test split; varying the random seed produced improvements in the range of approximately 55–66%, suggesting the precise value should be interpreted with some caution, though the direction and magnitude of the improvement remain consistent. Furthermore, this evaluation is in-sample with respect to time: the test locations fall within July 1993, so the kriging predictor can exploit temporally nearby training observations. Performance on out-of-sample time periods would likely be lower.

Overall, kriging is clearly the more realistic predictor for this dataset. Real temperature fields are spatially and temporally correlated in ways that a simple linear trend cannot capture. The OLS surface is overly smooth and fails to reflect local weather patterns, while kriging produces surfaces that are both consistent with the large-scale trend and responsive to local observations.

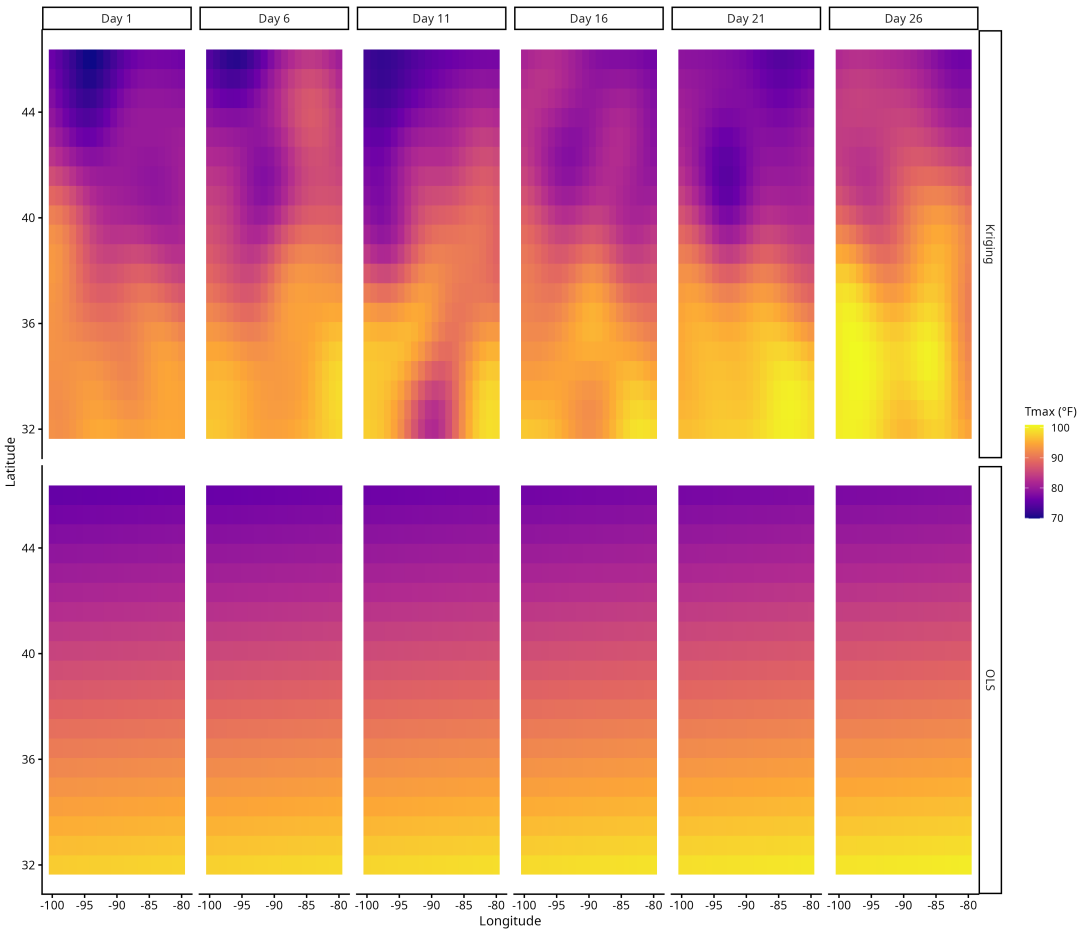

Figure 5: Comparison of OLS (bottom) and universal kriging predictions (top) of Tmax across the spatial grid for six days in July 1993.

4. Discussion and Conclusion

The comparison between OLS and universal kriging shows a clear and substantial performance gap in favour of kriging. Quantitatively, kriging achieves a 66.3% reduction in RMSE, indicating that modelling the spatio-temporal covariance structure leads to a major improvement rather than a marginal refinement. Qualitatively, OLS produces smooth, nearly identical surfaces across time, whereas kriging captures local variation and evolving spatial patterns that better reflect realistic temperature dynamics. The covariance model is therefore central to predictive performance.

Kriging performs best when prediction locations are close in space and time to observed data. In these cases, the covariance-weighted residual correction term incorporates strong local information and adapts predictions to fine-scale structure. However, its advantage diminishes in regions with sparse station coverage or for time points far from observed days, where the covariance vector approaches zero and predictions revert to the trend, behaving similarly to OLS. The short estimated temporal lengthscale further limits the influence of observations across time.

Both approaches have limitations. OLS assumes independent errors, which is violated in this spatio-temporal setting, and is restricted to a global planar trend that cannot capture local structure. Kriging addresses these issues but introduces others: covariance parameters were estimated on a subset of 2000 observations for computational

reasons, which may limit representativeness; the Gaussian covariance function imposes a specific smoothness assumption; evaluation is in-sample with respect to time and likely overestimates performance; and the method is computationally expensive due to large matrix inversions.

Several improvements are possible. Covariance estimation could be performed on the full dataset using sparse or approximate methods. Alternative covariance families, such as the Matérn class, may provide more flexible modelling of smoothness. Spatial anisotropy could be extended to include rotation, and evaluation should be performed on temporally held-out data to better assess generalisation. Finally, non-separable spatio-temporal covariance functions could capture more complex dependence structures, such as the interaction between space and time that arises when weather systems propagate across a region (Wikle, Zammit-Mangion, and Cressie, 2019, p.143).

In summary, explicitly modelling spatio-temporal dependence is crucial for this type of data. OLS provides a simple baseline capturing large-scale trends but fails to represent local and dynamic structure. Universal kriging produces more accurate and realistic predictions by incorporating covariance information, at the cost of increased model complexity and computational demand.

Generative AI Statement

The use of Generative Artificial Intelligence (GenAI) was limited to debugging IDE error messages, and looking up R functions.

References

Wikle, C., Zammit-Mangion, A., and Cressie, N., 2019. *Spatio-temporal statistics with r* [Online], Chapman & hall/crc the r series. CRC Press. Available from: https://books.google.co.uk/books?id=FD-IDwAAQBAJ.